

# PrenticeMath Formula Reference Sheet

## Algebra & Coordinate Geometry

| Concept              | Formula                                    | Notes / Context                                  |
|----------------------|--------------------------------------------|--------------------------------------------------|
| Slope Formula        | $m = (y_2 - y_1) / (x_2 - x_1)$            | Finds steepness between two coordinate points    |
| Slope-Intercept Form | $y = mx + b$                               | $m$ = slope, $b$ = y-intercept                   |
| Point-Slope Form     | $y - y_1 = m(x - x_1)$                     | Useful when given a slope and one point          |
| Quadratic Formula    | $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$       | Solves $ax^2 + bx + c = 0$                       |
| Distance Formula     | $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ | Calculates straight-line distance between points |

## Geometry & Area

| Shape     | Area Formula                     | Perimeter / Circumference                           |
|-----------|----------------------------------|-----------------------------------------------------|
| Rectangle | $A = l \cdot w$                  | $P = 2l + 2w$                                       |
| Triangle  | $A = \frac{1}{2}b \cdot h$       | $P = \text{side}_1 + \text{side}_2 + \text{side}_3$ |
| Circle    | $A = \pi r^2$                    | $C = 2\pi r$ (or $C = \pi d$ )                      |
| Trapezoid | $A = \frac{1}{2}(a + b) \cdot h$ | Sum of all four outer sides                         |

## Trigonometry & Right Triangles

| Concept                         | Ratio / Identity                                    | Memory Hook / Notes                         |
|---------------------------------|-----------------------------------------------------|---------------------------------------------|
| Pythagorean Theorem             | $a^2 + b^2 = c^2$                                   | Right triangles only; $c$ is the hypotenuse |
| Sine Ratio ( $\sin \theta$ )    | $\sin \theta = \text{Opposite} / \text{Hypotenuse}$ | SOH - Part of SOH-CAH-TOA                   |
| Cosine Ratio ( $\cos \theta$ )  | $\cos \theta = \text{Adjacent} / \text{Hypotenuse}$ | CAH - Part of SOH-CAH-TOA                   |
| Tangent Ratio ( $\tan \theta$ ) | $\tan \theta = \text{Opposite} / \text{Adjacent}$   | TOA - Part of SOH-CAH-TOA                   |

*This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.*